

Chapter 9 Network

A note on the

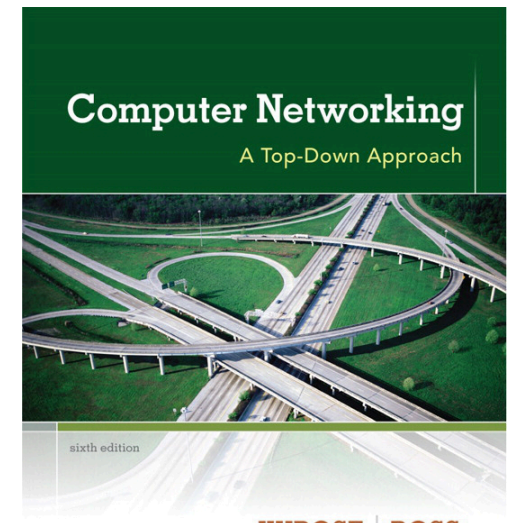
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*orking: A
Top Down
Approach
6th edition
Jim Kurose, Keith
Ross
Addison-Wesley
March 2012*

Chapter 9: Network Management

Chapter goals:

- ❖ introduction to network management
 - motivation
 - MIB: management information base
 - SMI: data definition language
 - SNMP: protocol for network management
 - security and administration
- ❖ presentation services: ASN.1

Chapter 9

~~outline~~

- ❖ What is network management?
- ❖ Internet-standard management

SM

- Management Information Base: MIB
- SNMP Protocol Operations and Transport Mappings
 - Security and Administration
- ❖ ASN.1

What is network management?

- ❖ **autonomous systems (aka “network”)**: 1000s of interacting hardware/software components
- ❖ other complex g monitoring,

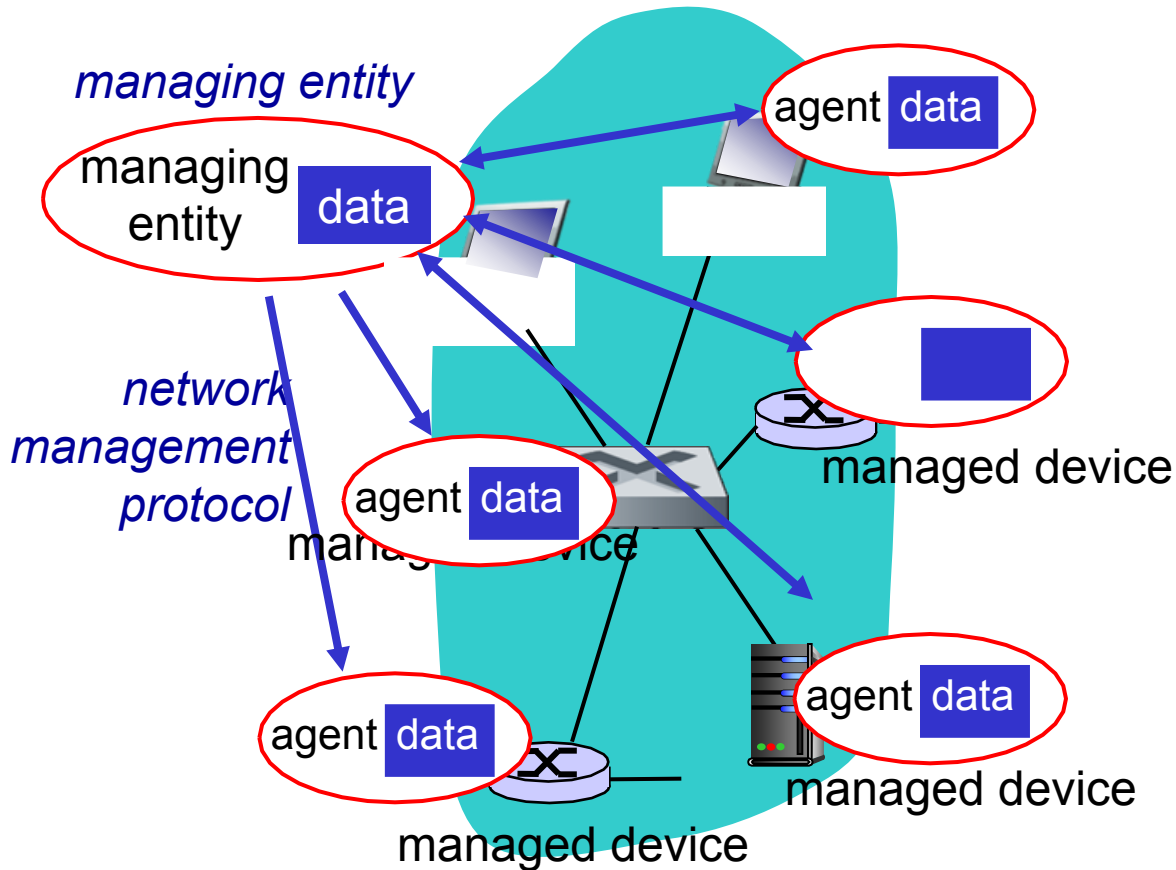
- nuclear power plant
- ~~others?~~



"**Network management** includes the deployment, integration and coordination of the hardware, software, and human elements to monitor, test, poll, configure, analyze, evaluate, and control the network and element resources to meet the real-time, operational performance, and Quality of Service requirements at a reasonable cost."

Infrastructure for network management

definitions:



managed devices
contain

thered into a
*Management
Information
Base (MIB)*

Network management standards

OSI CMIP

- ❖ Common

Proto

- ❖ designed 1980's:
the unifying net management standard
- ❖ too slowly standardized

SNMP: Simple Network Management col

- ❖ started simple
- ❖ deployed, adopted rapidly
- ❖ growth: size, complexity
- ❖ currently: **SNMP V3**

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SNMP overview: 4 key parts

- ❖ **Management information base (MIB):**
 - distributed information store of network management
- ❖ **(SMI)**
 - data definition language for MIB objects
- ❖ **SNMP protocol**
 - convey manager<->managed object info, commands
- ❖ **security, administration capabilities**
 - major addition in SNMPv3

SMI: data definition language

Purpose: syntax,
semantics of
management

Basic Data Types

INTEGER
Integer32

- ❖ base data types:
 - straightforward,
boring
- ❖ OBJECT-TYPE
 - data type, status,
semantics of managed
object
- ❖ MODULE-IDENTITY

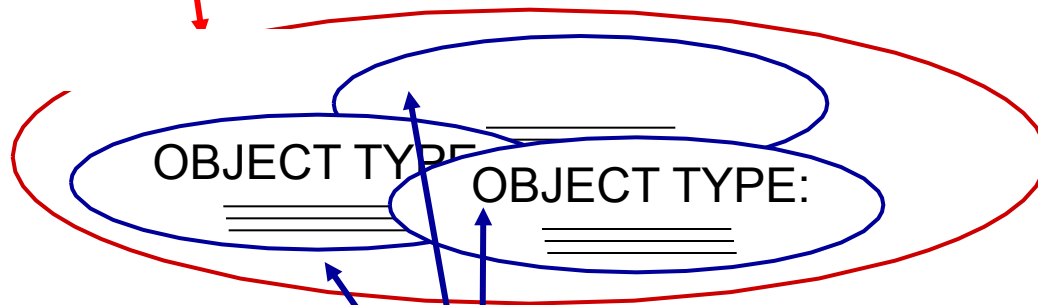
IFIED
IPAddress
Counter32
Counter64
Gauge32
Time Ticks
Opaque

SNMP MIB

MIB module specified via SMI

MODULE-IDENTITY

(100 standardized MIBs, more vendor-specific)



objects specified via SMI
OBJECT-TYPE construct

SMI: object, module examples

OBJECT-TYPE:

ipInDelivers

ipInDelivers OBJECT-TYPE

STATUS current

DESCRIPTION

“The total number of input datagrams successfully delivered to IP user-protocols (including ICMP)”

::= { ip 9 }

MODULE-IDENTITY:

ipMIB

ipMIB MODULE-IDENTITY

DATED “941101000Z”

.....”

DESCRIPTION

“The MIB module for managing IP and ICMP implementations, but excluding their management of IP routes.”

REVISION “019331000Z”

.....

::= { mib-2 48 }

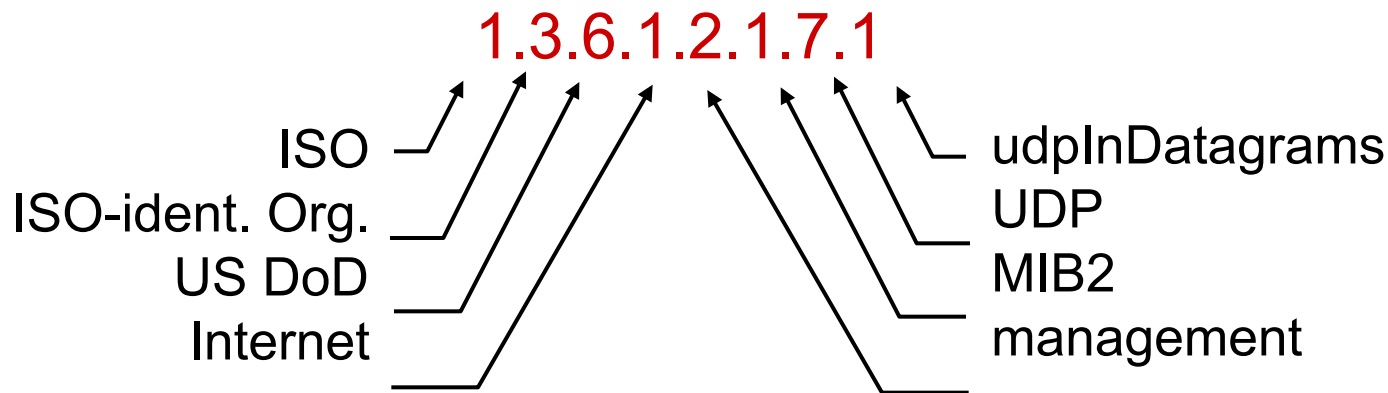
MIB example: UDP module

<u>Object ID</u>	<u>Name</u>	<u>Type</u>	<u>Comments</u>
1.3.6.1.2.1.7.1	UDPInDatagrams	Counter32	total # datagrams delivered at this node
1.			
1.3.6.1.2.1.7.			datagrams: all other reasons
1.3.6.1.2.1.7.4	UDPOutDatagrams	Counter32	# datagrams sent
1.3.6.1.2.1.7.5	udpTable	SEQUENCE	one entry for each port in use by app, gives port # and IP address

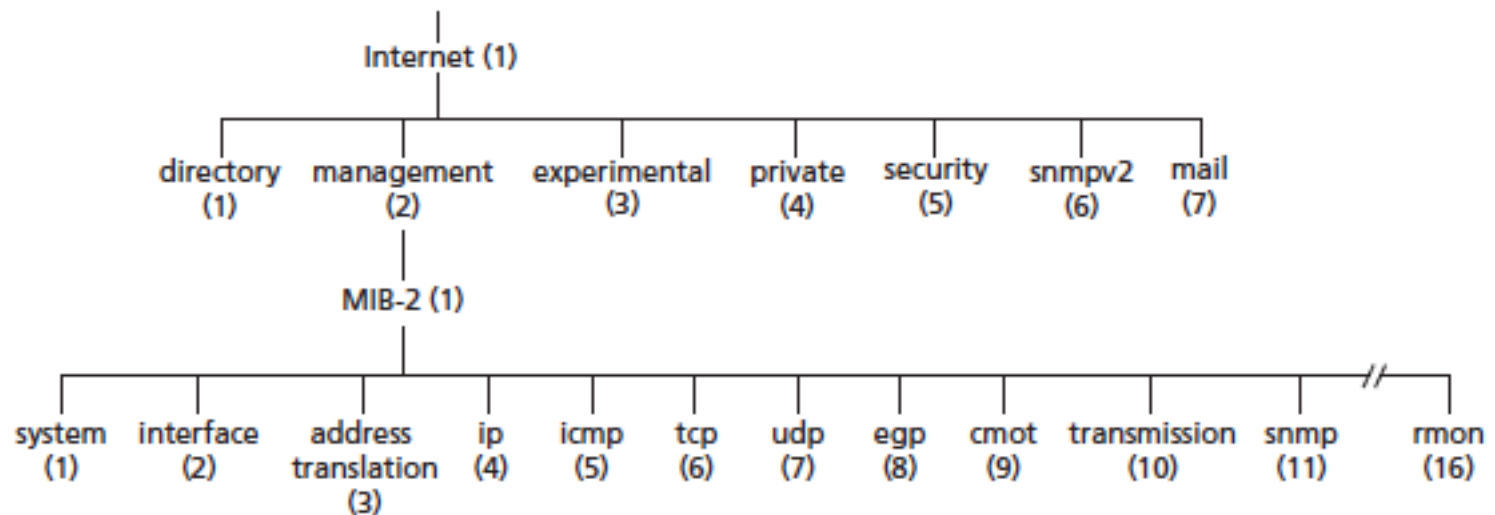
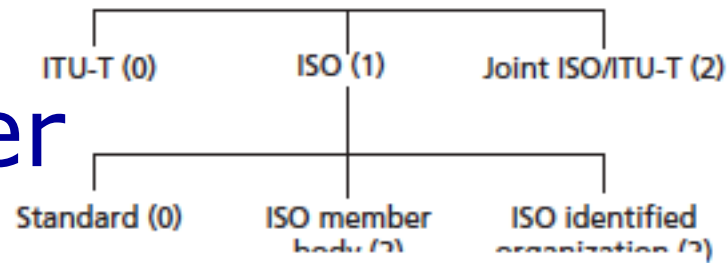
SNMP naming

question: how to name every possible standard object (protocol, data, more..) in every pos standard??

- hi s
- each branchpoint has name, number

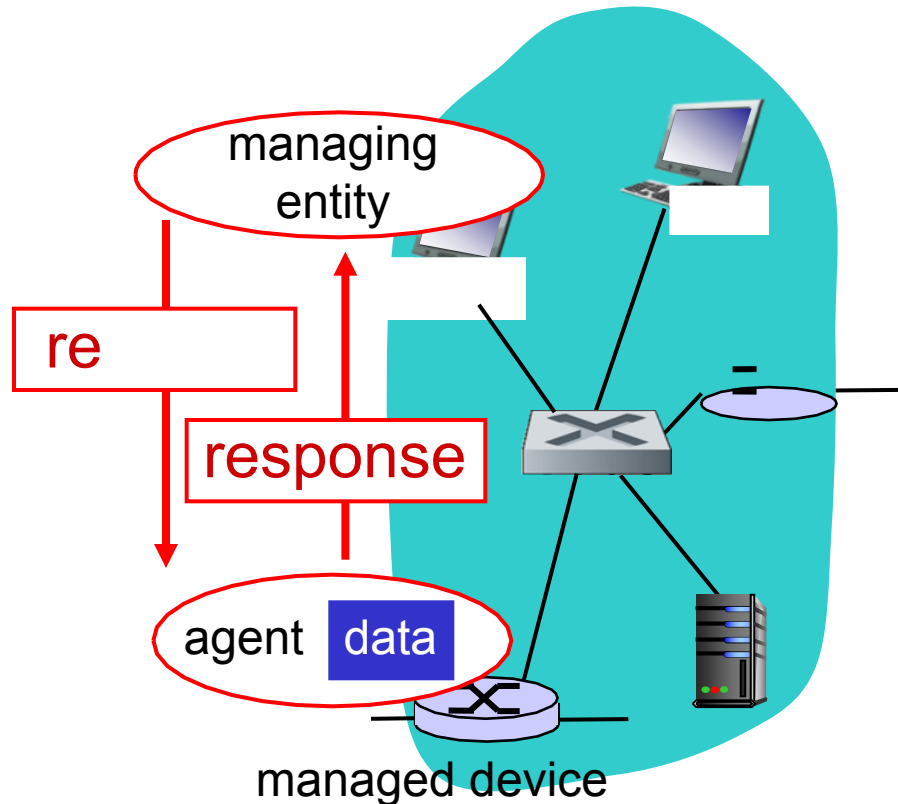


OSI Object Identifier Tree

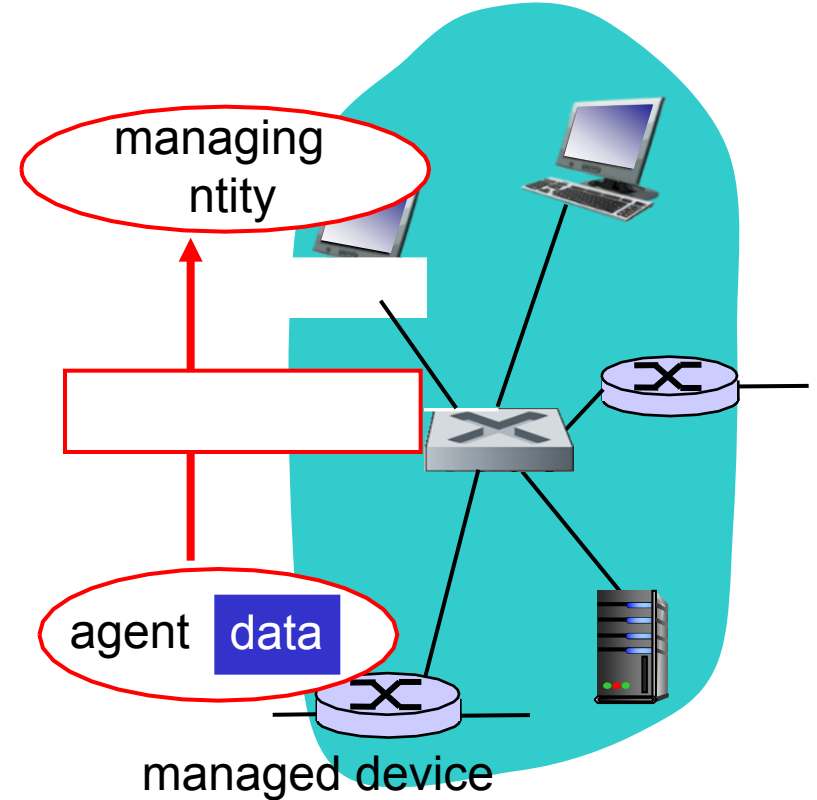


SNMP protocol

Two ways to convey MIB info, commands:



request/response mode

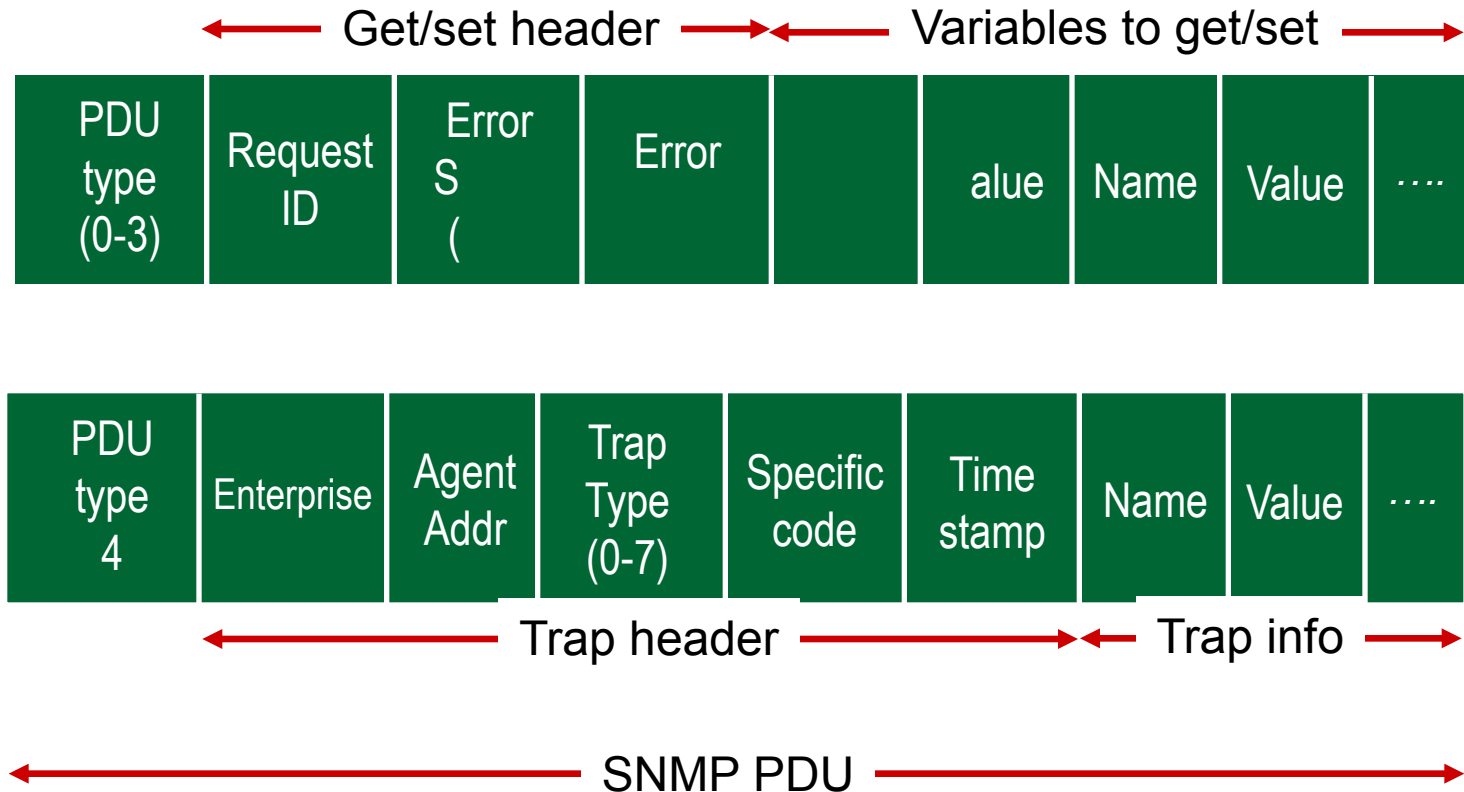


trap mode

SNMP protocol: message types

<u>Message type</u>	<u>Function</u>
GetRequest GetNextRequest GetBulkRequest	Mgr-to-agent: “get me data” list, block)
SetRequest	Mgr-to-agent: set MIB value
Response	Agent-to-mgr: value, response to Request
Trap	Agent-to-mgr: inform manager of exceptional event

SNMP protocol: message formats



SNMP security and administration

- ❖ **encryption:** DES-encrypt SNMP message
- ❖ **authentication** end
- ❖ **protection against playback:** use nonce
- ❖ **view-based access control:**
 - SNMP entity maintains database of access rights, policies for various users
 - database itself accessible as managed object!

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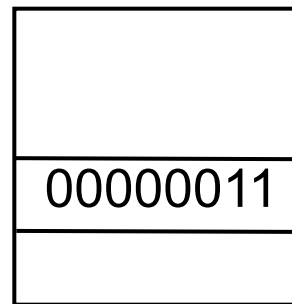
- Management Information Base: MIB
- SNMP Protocol Operations and Transport Mappings
- Security and Administration
- ❖ The presentation problem: ASN.1

The presentation problem

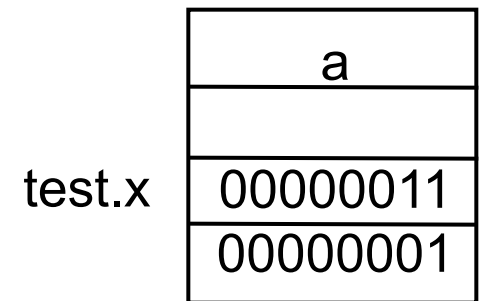
Q: does perfect memory-to-memory copy solve “the communication problem”?

A: not always!

```
char c  
int x;  
} test;  
test.x = 256;  
test.code='a'
```



host 1 format



host 2 format

problem: different data format, storage conventions

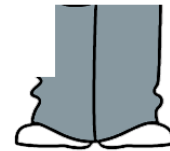
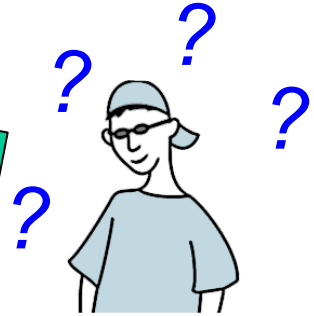
A real-life presentation problem:



grandma



aging 60's
hippie



2012 teenager

Presentation problem: potential solutions

1. Sender learns receiver's format. Sender translates into receiver's format. Sender sends.

– real-world

2. Send sender's format. Receiver translate into receiver-local format

– real-world-analogy

– pros and cons?

3. Sender translates host-independent format. Sends. Receiver translates to receiver-local format.

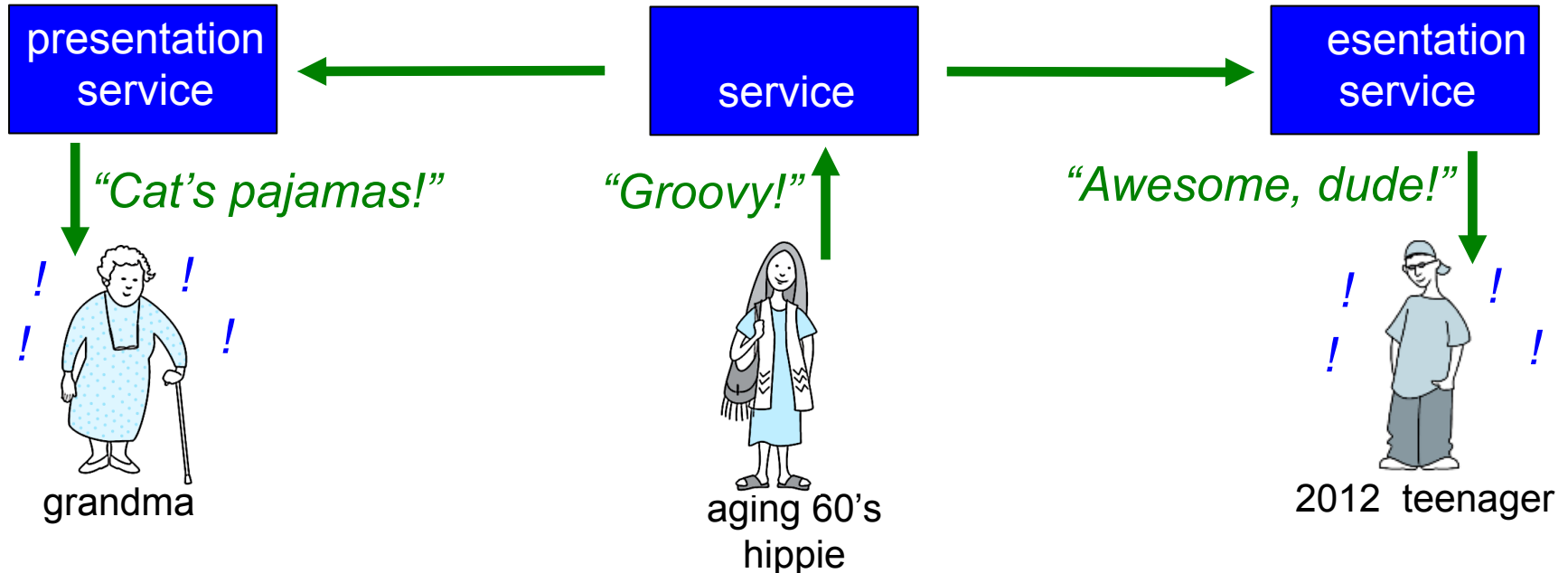
– real-world analogy?

pros and cons?

Solving the presentation problem

1. Translate local-host format to host-independent format
2. Transmit data in host-independent format
- 3

t



ASN.1: Abstract Syntax Notation 1

❖ ISO standard X.680

- used extensively in Internet
- like eating venging this “good for

- like SMI

❖ BER: Basic Encoding Rules

- specify how ASN.1-defined data objects to be transmitted
- each transmitted object has Type, Length, Value (TLV) encoding

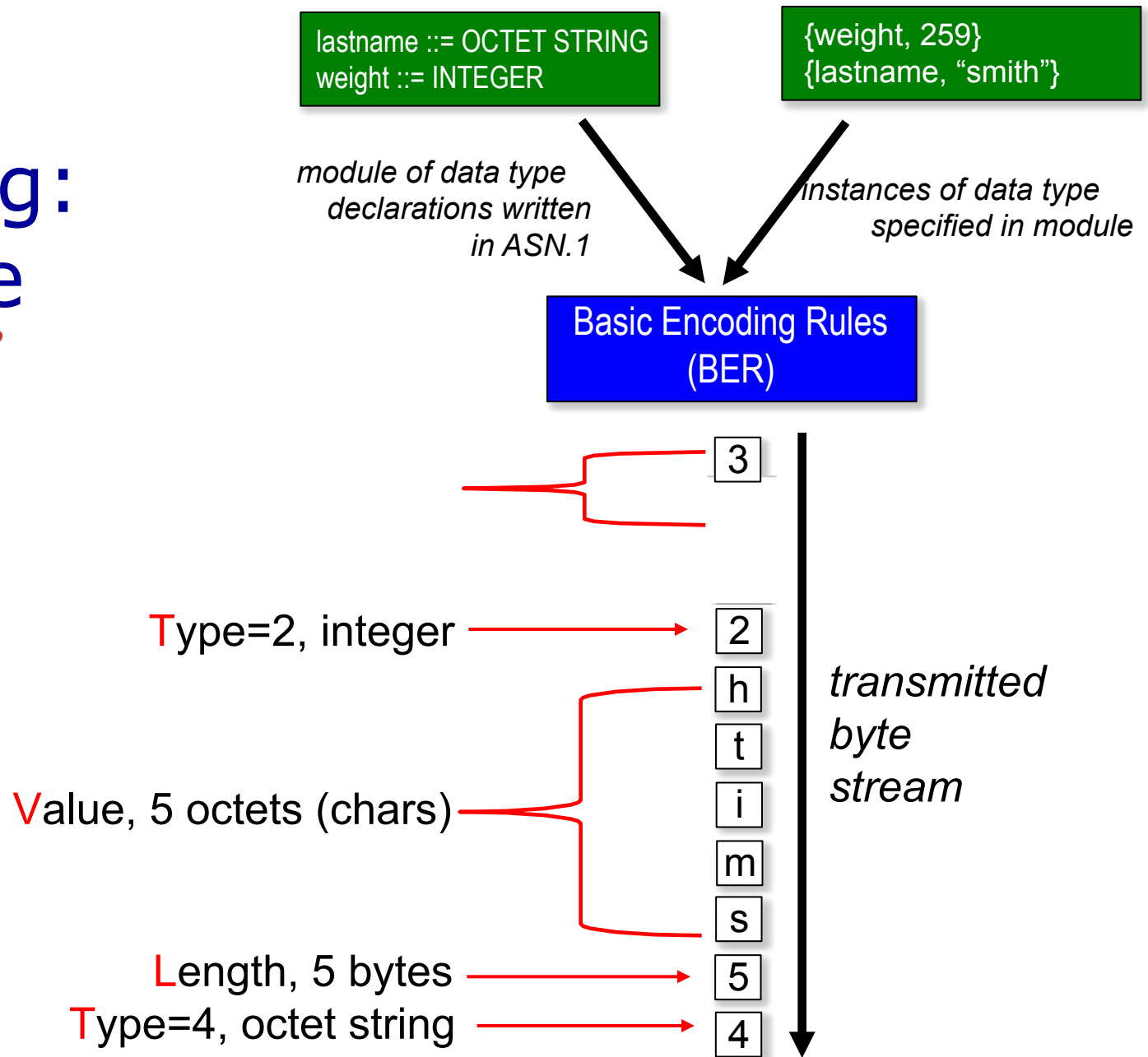
TLV Encoding

Idea: transmitted data is self-identifying

- I: data type, one of ASN.1-defined types
- L: length of

1	Boolean
2	Integer
3	Bitstring
4	Octet string
5	Null
6	Object Identifier
9	Real

TLV encoding: example



Network management: summary

❖ network management

- extremely important: 80% of network “cost”
- ASN.I for dat

info

❖ network management: more art than science

- what to measure/monitor
- how to respond to failures?
- alarm correlation/filtering?